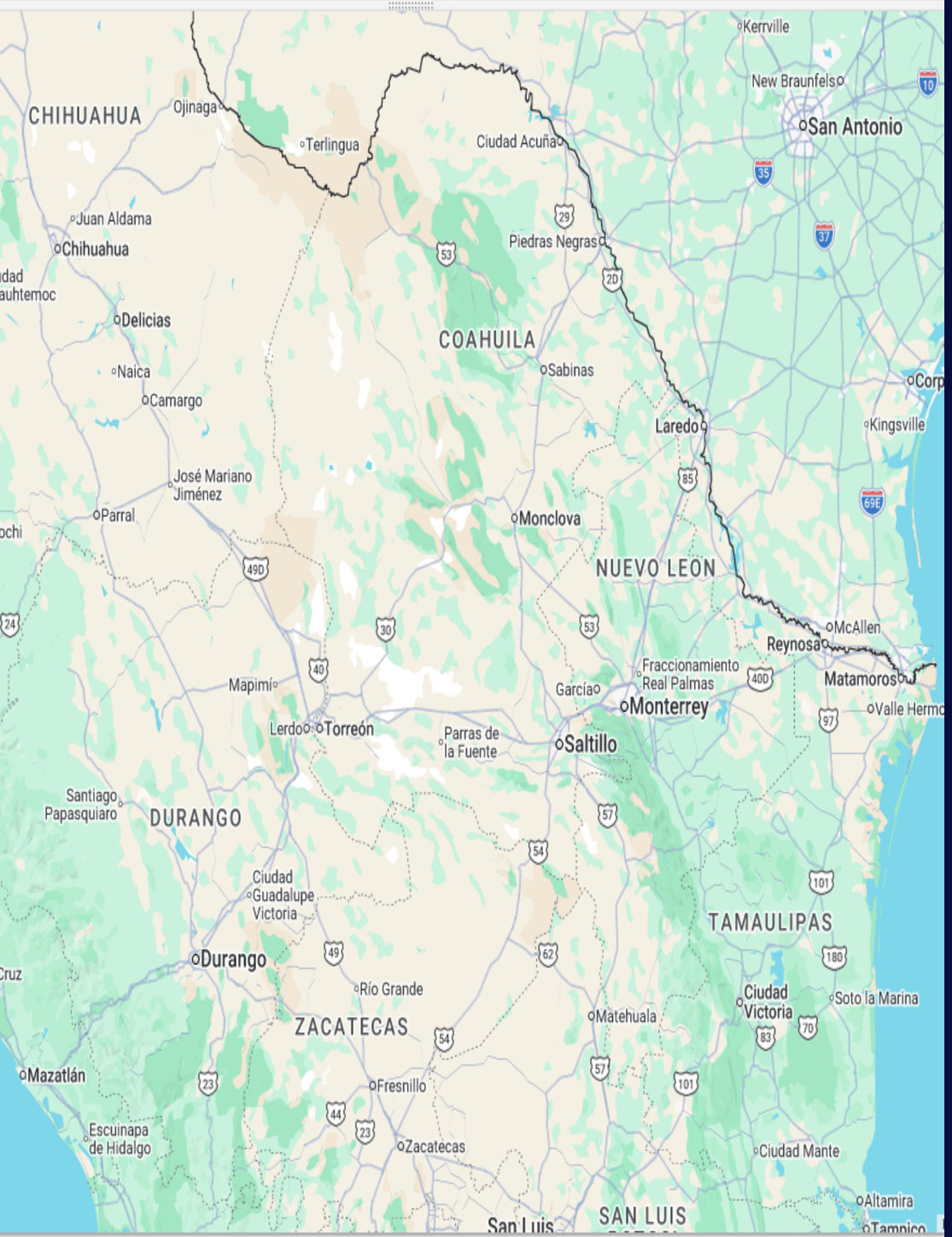

Detecting and Modeling Spatio-Temporal Environmental Stress Patterns

Using Multi-Spectral Satellite Imagery to Support Water Equity Decision-Making

Background & Motivation

The Water Inequity Challenge

Water inequity in underserved communities is driven by complex environmental stressors that traditional monitoring methods struggle to capture. Prolonged drought, infrastructure degradation, seasonal environmental stress, land cover change, vegetation decline, water contamination patterns, sedimentation, and hydrological disruption all contribute to unequal water access.

Current monitoring relies on precipitation indices (SPI), hydrological station data, manual inspection, and static policy rules. However, these methods lack spatial granularity, do not integrate multi-source environmental signals, fail to model temporal anomalies, and are not scalable for large regions.

The Satellite Solution

Satellite imagery provides continuous spatial coverage, multi-spectral information beyond RGB, long historical archives, and opportunity for automated large-scale analysis. This project proposes a computer vision framework that goes beyond binary classification and instead models environmental stress dynamics over space and time.

Research Problem

1

Traditional Question

"Is this region drought or not?"

2

Our Approach

Modeling environmental stress evolution

Instead of simple binary classification, we ask fundamental questions about environmental stress dynamics:

Temporal Evolution

How does environmental stress evolve over time and what are the seasonal patterns?

Anomaly Detection

Can we detect abnormal water stress patterns that indicate emerging problems?

Hidden Factors

Can multi-spectral signals reveal hidden stress factors invisible to traditional methods?

Equity Mapping

Can we identify regions experiencing inequitable environmental burden?

Policy Support

Can we quantify environmental stress indicators to support policy decisions?

Core Research Question

Can multi-spectral satellite imagery be used to detect and model spatio-temporal environmental stress patterns that provide actionable evidence for equitable water distribution strategies?

Project Objectives

01

Build Multi-Spectral Pipeline

Develop a satellite image processing pipeline that handles multi-spectral data from multiple sources

02

Extract Environmental Indicators

Derive interpretable environmental indicators from satellite data including vegetation, water, and soil indices

03

Model Spatio-Temporal Changes

Model how environmental stress changes across both space and time dimensions

04

Detect Anomalies

Identify abnormal patterns and long-term degradation trends in environmental stress

05

Translate to Equity Metrics

Convert environmental stress indicators into water equity decision metrics

Data Sources

Multi-spectral datasets provide information beyond RGB, enabling comprehensive environmental monitoring:

Sentinel-2 (ESA)

High-resolution multi-spectral imagery with frequent revisit time

Landsat 8/9

Long historical archive with consistent spectral bands

Google Earth Engine

Cloud-based platform for large-scale satellite data processing

MODIS

Moderate resolution with daily global coverage

Copernicus Hub

Open access to European satellite data

Spectral Bands Used

- Red, Green, Blue (RGB)
- Near Infrared (NIR)
- Short-Wave Infrared (SWIR)
- Thermal bands (when available)

Temporal Coverage

Minimum 5–10 years of seasonal data with monthly or quarterly sampling frequency

Methodology

Data Preprocessing

- **Atmospheric Correction**
  Remove atmospheric effects from satellite measurements
- **Cloud Masking**
  Identify and filter cloudy pixels
- **Geospatial Alignment**
  Ensure consistent spatial referencing
- **ROI Selection**
  Define regions of interest
- **Temporal Alignment**
  Match timestamps across different sensors

Feature Engineering

Instead of raw RGB classification, we extract meaningful environmental indices:

Vegetation Indices

NDVI (Normalized Difference Vegetation Index) and EVI (Enhanced Vegetation Index) for plant health assessment

Water Indices

NDWI (Normalized Difference Water Index) for water content detection

Soil Indicators

NDBI and surface reflectance metrics for dryness assessment

Texture Features

GLCM and local binary patterns for spatial pattern analysis

Temporal Metrics

Seasonal variance, trend slopes, and deviation from historical baseline

Spatio-Temporal Modeling Framework

Three layered modeling approach captures both spatial patterns and temporal dynamics:

Layer 1: Static Classification

Drought severity classification using Random Forest, SVM, or CNN for image-based approaches

Layer 2: Temporal Modeling

Detect changes across time using LSTM networks, Temporal CNN, change detection algorithms, and seasonal decomposition

Layer 3: Anomaly Detection

Identify abnormal stress patterns using Autoencoders, Isolation Forest, and Z-score anomaly detection methods

System Architecture & Mathematical Formulation

Processing Pipeline

1. Multi-spectral satellite images
2. Image preprocessing
3. Feature extraction
4. Temporal modeling
5. Stress scoring
6. Anomaly detection
7. Visualization

Mathematical Formulation

Let x represent spatial units (regions or grid cells) and $f_{\text{stress}}(x, t)$ represent environmental stress index over time.

Environmental stress modeled as:

$$S(x, t) = w_1 \quad \text{NDVI}(x, t) + w_2 \quad \text{NDWI}(x, t) + w_3 \quad \text{Texture}(x, t) + w_4 \quad \Delta\text{Season}(x, t)$$

Where ΔSeason measures deviation from historical seasonal baseline.

Anomaly detection:

$$A(x, t) = \quad | \; S(x, t) - \mu_{\text{season}}(x) \; |$$

High $A(x, t)$ indicates abnormal stress.

Water equity priority score:

$$W_{\text{priority}}(x) = \alpha \quad A(x, t) + \beta \quad \text{Vulnerability}(x)$$

Where $\text{Vulnerability}(x)$ includes socio-economic factors.

Outputs

Environmental Stress Map
Spatial visualization of stress patterns

Risk Zones
Identified high-stress regions

Stress Trend Graph
Temporal evolution visualization

Water Priority Ranking
Equity-based distribution recommendations

Expected Contributions & Research Significance

Multi-Spectral Framework
A comprehensive environmental stress modeling framework using multi-spectral satellite data

Temporal Anomaly Pipeline
A temporal anomaly detection pipeline that identifies abnormal stress patterns

Equity Integration
Integration of environmental signals with equity metrics for policy support

Scalable Approach
A scalable computer vision approach beyond binary drought detection

Why This Is Research-Level

Unlike simple drought classification systems, this project represents advanced research through:

- **Multi-Spectral Data Integration**
  Uses information beyond RGB to capture comprehensive environmental signals
- **Temporal Evolution Modeling**
  Models how stress changes over time rather than single-point snapshots
- **Anomaly Detection**
  Detects abnormal patterns that indicate emerging problems
- **Policy Connection**
  Connects environmental monitoring directly to policy decisions
- **Interdisciplinary Integration**
  Combines computer vision, machine learning, and temporal modeling
- **Decision-Support System**
  Moves toward operational system for water equity planning